

Computer vision

Image registration

Vanda Benešová

Image registration

Determining the **mapping** between two images of the same object, similar objects, the same region or similar regions.

Image registration is the process of **determining the spatial transform** that maps points from one image to corresponding points in the second image.

Application of image registration

- **Stitching** photos together, panoramas
- **Object comparison**, recognition
- **Object motion tracking**, human motion analysis – temporal alignment of prototypical postures, e.g. gesture understanding
- **Stereo reconstruction** and shape from motion – correspondence between key point, custom camera calibration
- **Video stabilization** – correspondence between key point
- **Medical imaging**

Image registration approaches

- Feature-based
- Image similarity measure

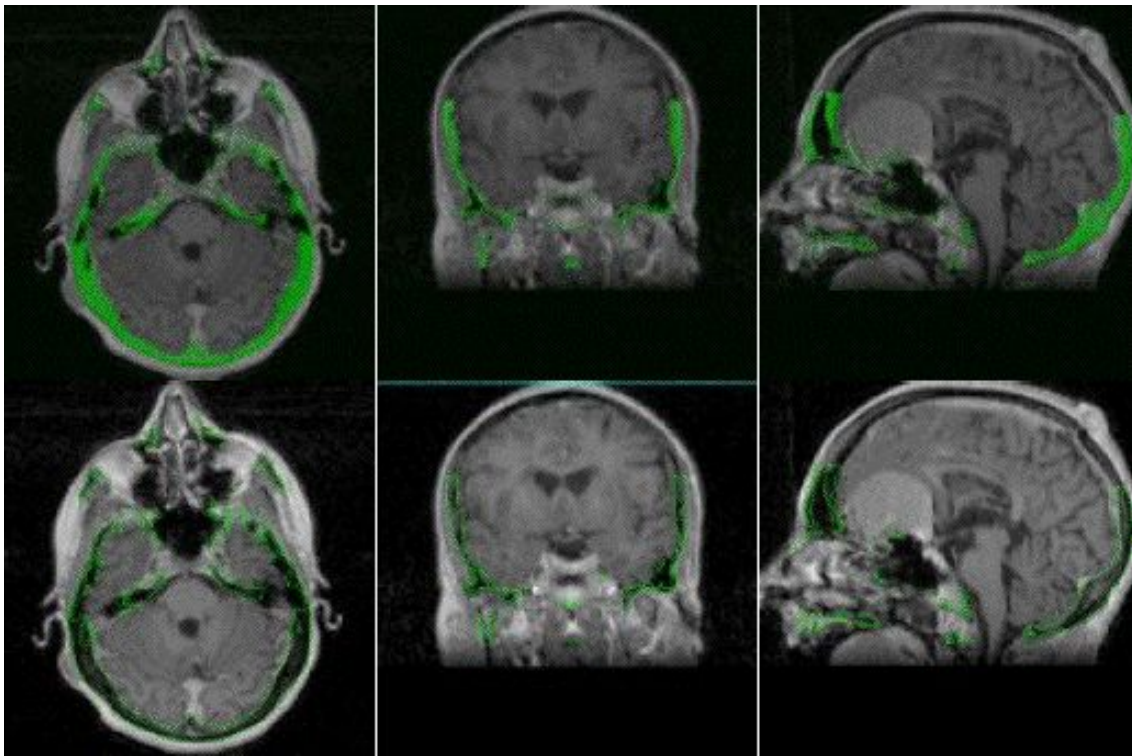
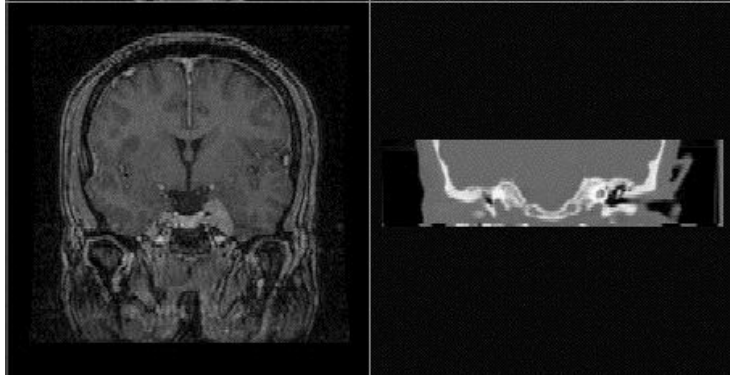
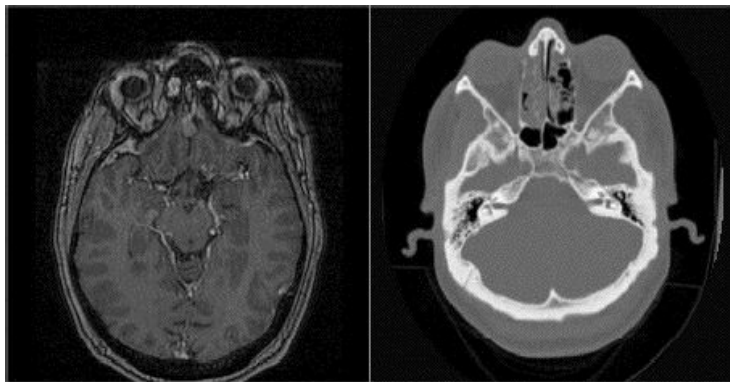
Types:

- Rigid (Affine...)
- Non rigid (Deformable registration..)

Example: MR-CT Brain Registration

MR CT

aligned images



Panorama - Stitching the images

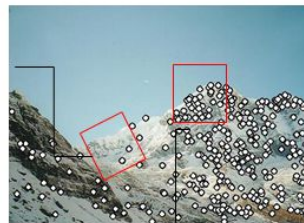
Detection: Identify the interest points

Description: Extract vector feature descriptor surrounding each interest point.

Matching: Determine correspondence between descriptors in two views



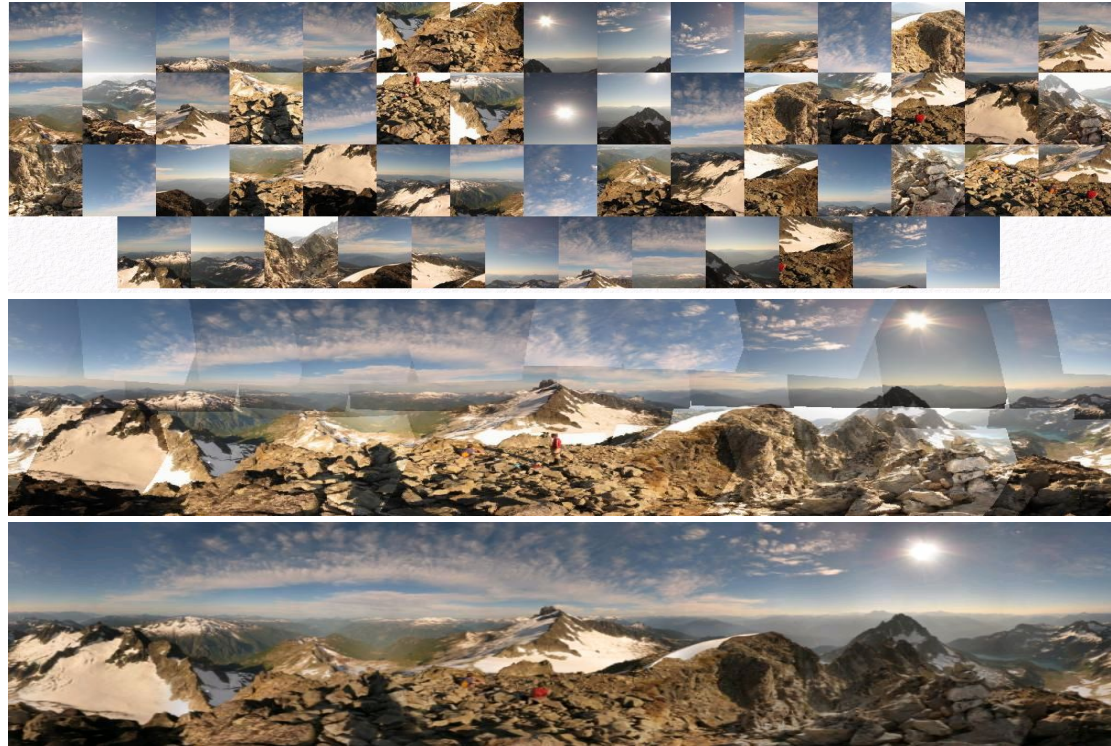
$$\mathbf{x}_1 = [x_1^{(1)}, \dots, x_d^{(1)}]$$



$$\mathbf{x}_2 = [x_1^{(2)}, \dots, x_d^{(2)}]$$



Automated mosaicing



Wide baseline stereo



[Image from T. Tuytelaars ECCV 2006 tutorial]

Fundamental task in medical imaging leads to rigid or non rigid registration

Typical use cases:

- Longitudinal studies, where temporal structural or anatomical changes are investigated.
- Matching of images from different patients.
- Multi-modal registration, i.e. matching images of the same patient acquired by different imaging technologies.

3 steps to perform image registration

1. Define how to measure the similarity
(also matching criteria, objective function) in the consequent optimization.
2. Get the deformation model.
 - Define the set of allowed transformations (correspondence functions, deformations)
3. Perform the optimization procedure.

Image similarity measure.

Mean squared error

Mean squared error $MSE = \sum (I_t(x, y) - I_s(x, y))^2$

Advantages:

- Simple and intuitive.
- Positive differences do not cancel negative differences.
- Ensures exact intensity matching.
-

Disadvantages:

- Sensitivity to outliers.
- Linear mean squares will not be the best fit for data that is not linear.
- Images must have almost the same brightness and contrast.
- Does not work if modalities differ.

Correlation coefficient

40/49

How can this template be found in the image?



Linear intensity changes are common in practice.

Mean squared error (MSE) may not work in such cases.
The correlation does.

Correlation coefficient

The (Pearson's) correlation coefficient ρ of the two random variables is the covariance of two variables divided (normalized) by their standard deviations σ .

In the case of images X (the reference image) and Y (the target image),

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Mutual information

The mutual information of two discrete random variables X and Y is based on the joint probability distribution $p(X, Y)$, which is approximated by the joint histogram in practice.

The mutual information is defined as

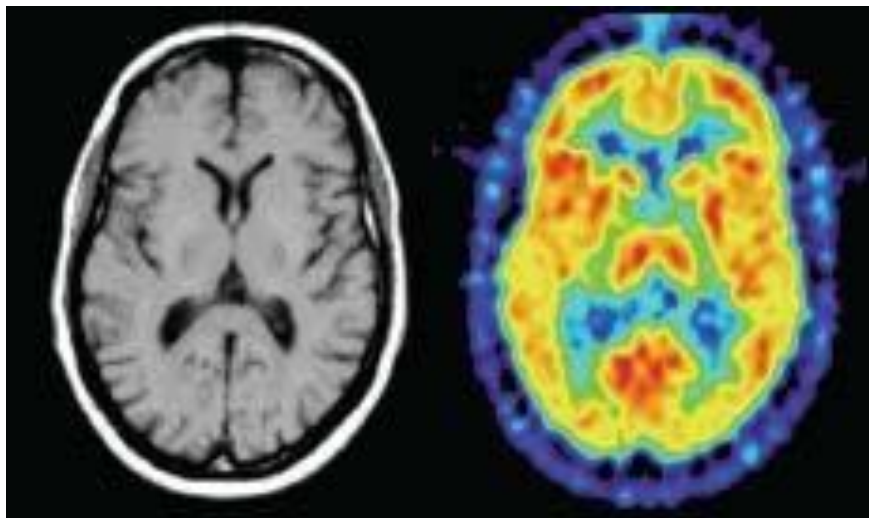
$$I(X; Y) = \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} P_{(X,Y)}(x, y) \log \left(\frac{P_{(X,Y)}(x, y)}{P_X(x) P_Y(y)} \right),$$

where $P(x)$, $P(y)$ are marginal probability density functions.

The mutual information measures the information that X and Y share. It tells how much knowing one of these variables reduces uncertainty about the other.

Mutual information example

Multimodal registration



Mutual information

Advantages:

- Rather general approach.
- Works for different modalities.

Disadvantages:

- Can be tricky to implement.
- Slower evaluation as compared to MSE and correlation.

Rigid / non rigid registration

Rigid vs. non-rigid registration

Rigid registration

- Landmark based
 - Edgels / surface based
- Pixel / voxel
 - intensity based
 - information theory based

Non-rigid registration

(also deformable registration)

- Registration using basis functions
- Registration using curves surfaces (e.g. splines)
- Physics based, e.g. elastic, fluid, optical flow, etc.

Rigid- non deformable registration

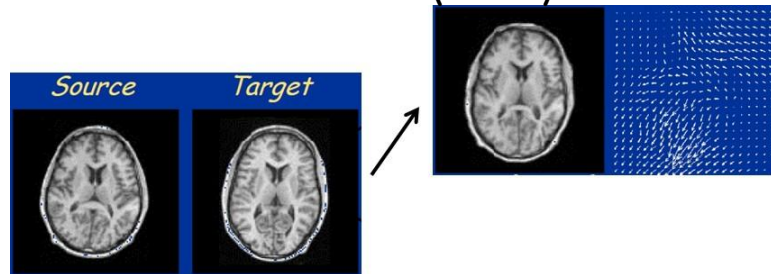
Intensity-based image registration methods

These are based on **pointwise/histogram** local comparisons between the reference and the target image. This is performed by a **sliding local window**.

Prior **preprocessing** of the images as denoising, edge filters, interest point filters can be used where appropriate.

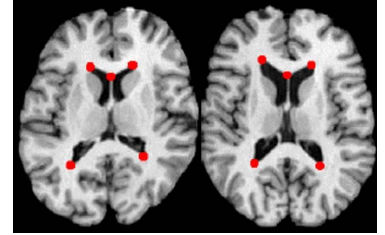
Correspondence is based on (di)similarity measures as

- Mean squared error (MSE) = Sum of squared differences (SSD). Dissimilarity.
- Normalized cross-correlation.
- Mutual information. Similarity.



Landmark- interactive based image registration methods

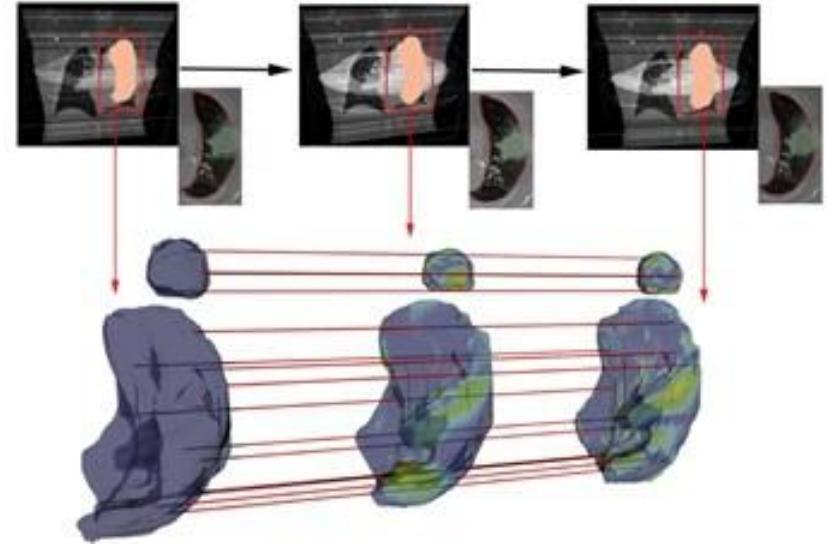
- Manually defined landmarks serve to infer the deformation
- Landmarks
 - Extrinsic landmarks, artificial objects attached to, e.g., a patient.
 - Intrinsic landmarks, e.g. obvious anatomical structures.
 - Translation is estimated from landmark centroids shifts.



Feature-based image registration methods

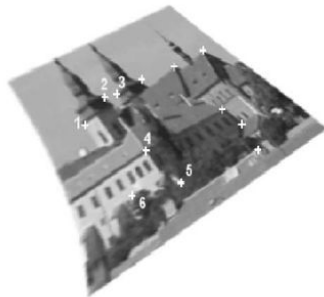
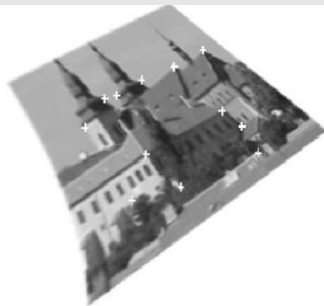
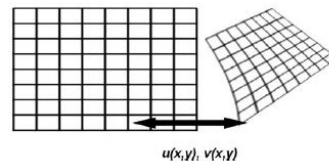
Find correspondences between the local features or regions, lines, contours.

SIFT, SURF, BRIEF...



Four steps of image registration

- feature detection
- feature matching by invariant descriptors (the corresponding pairs are marked by numbers).
- calculate Homography matrix (corresponding feature pairs + RANSAC)
- apply the Homography matrix
- image resampling and transformation using appropriate interpolation technique.



The goal of methods of image registration

It is assumed that two images are given:

the reference image $I_a(x, y)$ and the target image $I_b(x, y)$,
where x, y are coordinates of pixels.

$\phi()$ is **spatial (geometric) transformation** -> Homography matrix for translation, affine, projective transformation

$g()$ is **intensity transformation**.

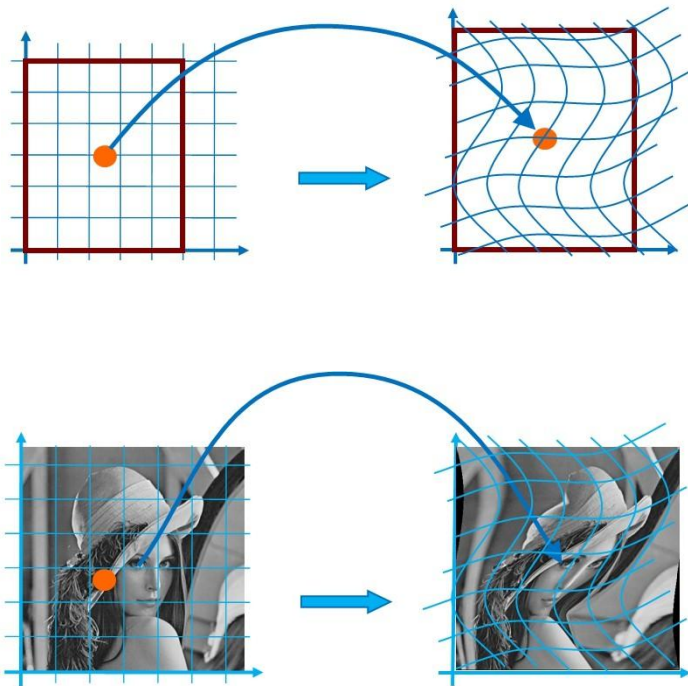
The goal is to find such $\phi()$ and $g()$ that the images are best matched

Then:

$$I_b(x, y) = g(I_a(\phi(x, y)))$$

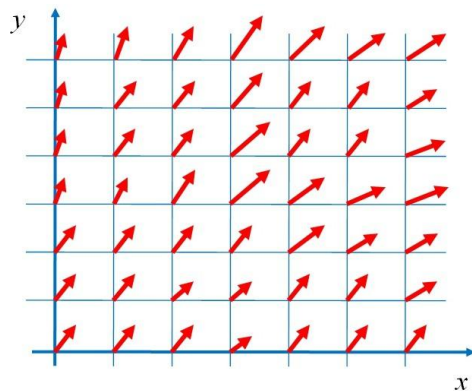
Non rigid - deformable registration

Deformation model - illustration

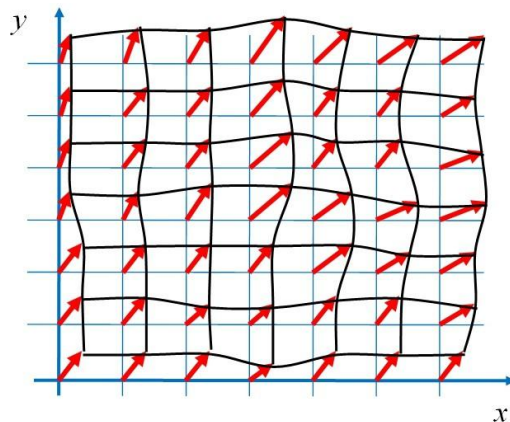


Deformation model visualisation

Estimated shifts per
correspondence



Estimated deformation field



Non-rigid (deformable) image registration

- Non-linear
- Many different parameterizations
- Too much flexibility in the transformation and its parameters can lead to undesirable results

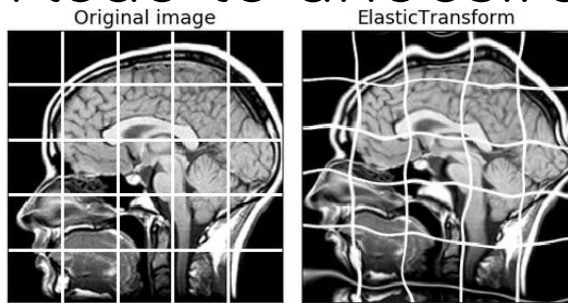
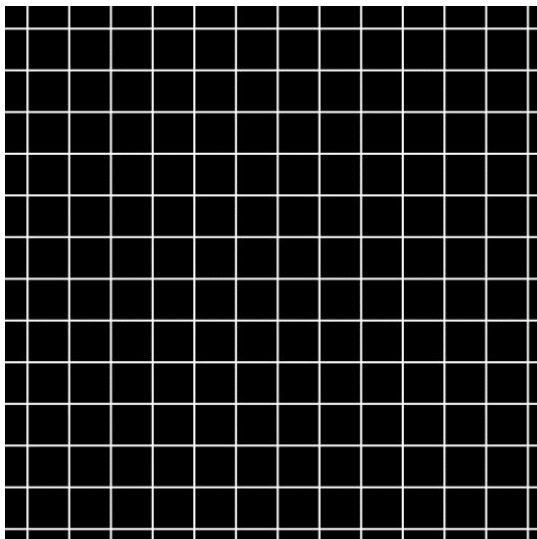


Image warping example - 0 %

Deformation grid



Brain image
warping

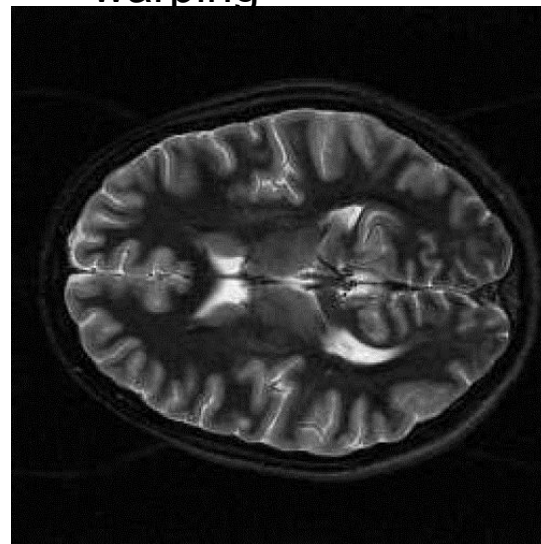
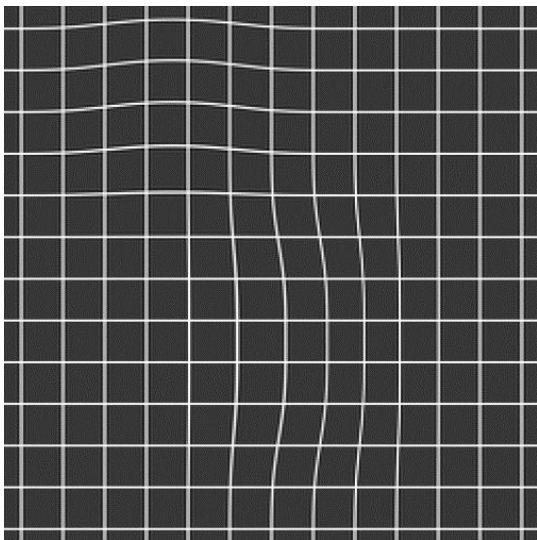


Image warping example - 25 %

Deformation grid



Brain image warping

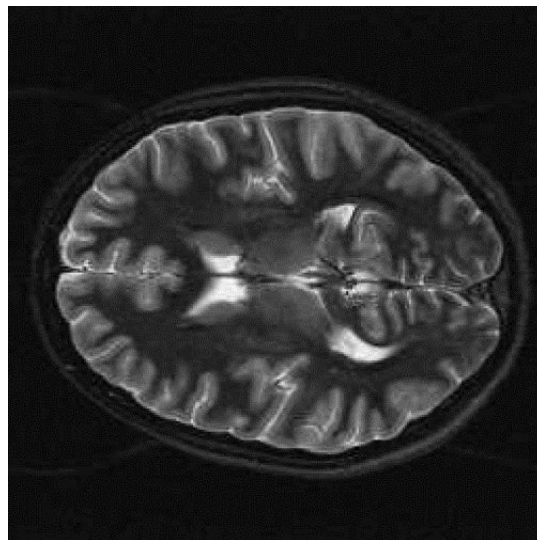
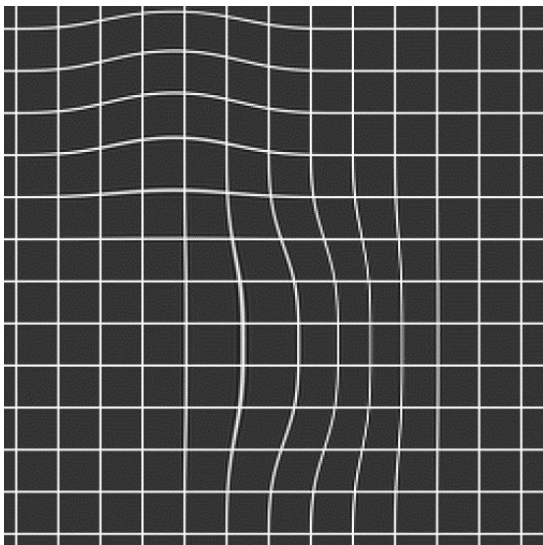


Image warping example - 50 %

Deformation grid



Brain image warping

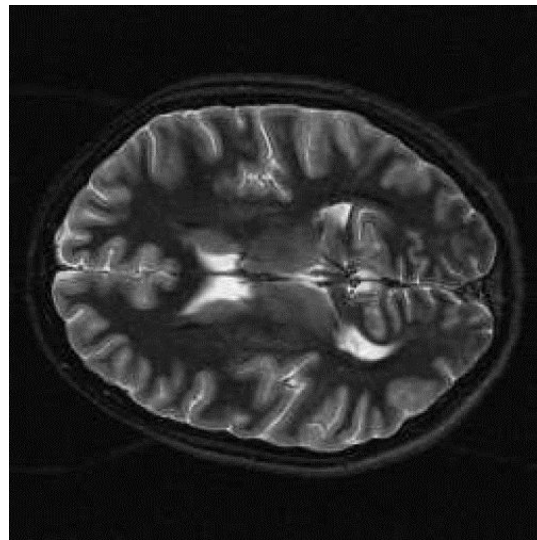
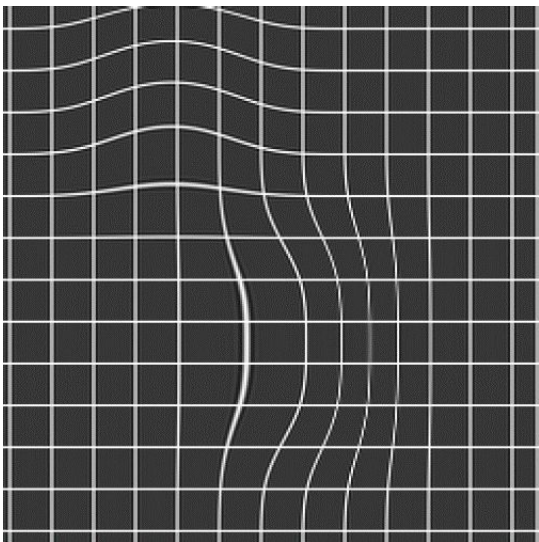


Image warping example - 75 %

Deformation grid



Brain image warping

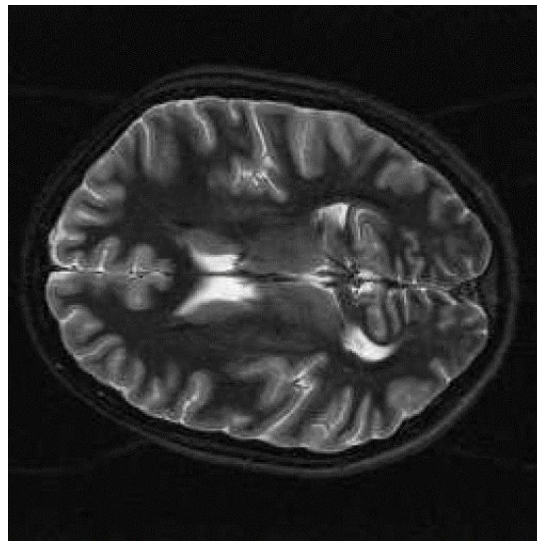
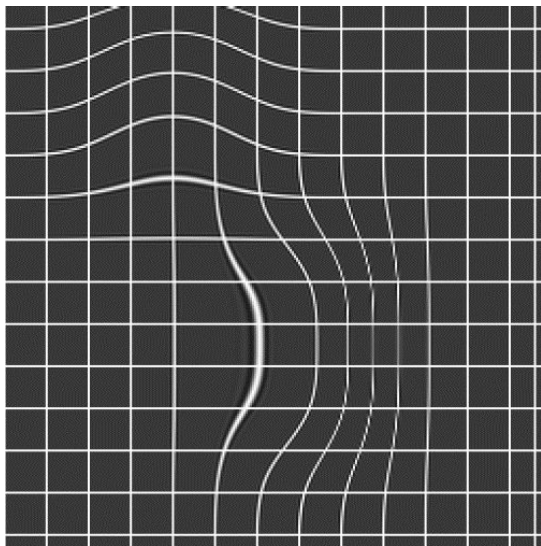
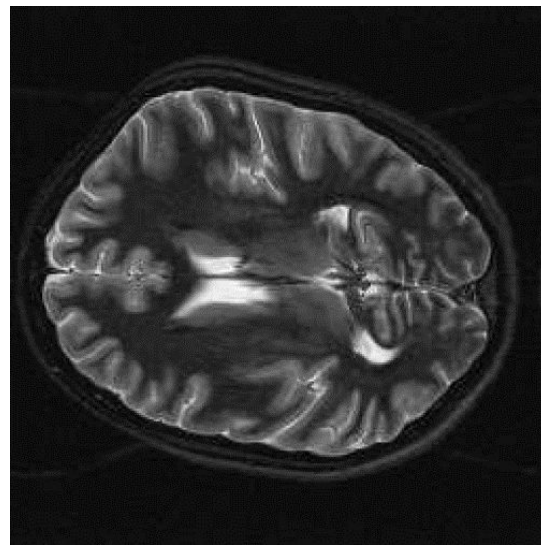


Image warping example - 100 %

Deformation grid



Brain image warping



Free-form deformation (FFD)

The general idea is to deform an image by manipulating a regular grid of control points that are distributed across the image at an arbitrary mesh resolution.

Control points can be moved and the position of individual pixels between the control points is approximated from the positions of surrounding control points.

